CHAPTER 7

Integration of GNSS and INS: Part 2

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7.1 Introduction

This chapter continues the examination of integrated GNSS/INS systems and deals with the "hands-on" aspect of GNSS/INS this integration. To this end we present three hands-on case studies. The data and MATLAB scripts required to reproduce the results presented and discussed in the chapter come with the software that accompanies this book. The readers are encouraged to follow along using these software examples.

The first case study will be a performance analysis of a low-cost, integrated GNSS/INS navigator designed as part of the navigation, guidance and control suite for a small UAV. This is followed by a case study of automotive application. This is an application where an integrated GNSS/INS system is used to estimate vehicle sideslip, which is an important parameter in many electronic stability control (ESC) systems. This case study will include a discussion and software demonstration of issues associated with state observability (particularly heading) in integrated GNSS/INS systems. These two case studies are applications where low-cost inertial sensors of automotive- or consumer-grade quality provide the requisite performance. In closing the chapter, however, we will switch gears slightly and focus on high-accuracy applications, which often use (although do not necessarily require) higher quality (navigation grade) inertial navigators. This final case study deals with the problem of enhancing the integer ambiguity resolution in carrier phase differential GNSS applications.

7.2 Case Study 1: Low-Cost GNSS/INS Integrated Navigator

As noted in Chapter 6, one area where the application of GNSS/INS integration using low-cost inertial sensors has been successful is in the area of guidance, navigation, and control suites for small aerial vehicles. In this application, the purpose of integrating the inertial sensors with GNSSs is primarily for providing a means of attitude determination using a single GNSS receiver/antenna. An added benefit of this integration is increased bandwidth of the position and velocity solution than would be available when using GNSSs alone. The attitude and higher bandwidth position and velocity solutions are used for various applications

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including automatic control of vehicles. This is possible because inertial sensors can easily provide measurements of angular velocity and specific force at rates in excess of 50 Hz.

Figure 7.1 is the ground trajectory of a small aerial vehicle that was equipped with inertial sensors of the automotive-grade quality and a GNSS receiver.[1] The vehicle is a small hand-launched flying-wing described in [2]. Figure 7.2 shows the output of the MATLAB-based GNSS/INS integration algorithm that comes with this book. These plots are a seven minute snippet from the time history for the attitude estimate generated by this system. The figure shows yaw (ψ_{nb}), pitch (θ_{nb}) and roll (ϕ_{nb}) estimates. The attitude histories are shown because with low-cost systems they are the harder states to estimate accurately. From Figure 7.2 we see that the yaw or heading angle is constantly changing. This is consistent with the ground track shown in Figure 7.1 (i.e., a continuous orbit about a point). Figure 7.3 shows the estimates of the accelerometer and rate gyro biases for the same time period shown in Figure 7.2. After the initial transient the estimates settle down to a steady state value. Note that the magnitude of the bias estimates is relatively large. For example, the gyro bias b_{g2} (i.e., output error on the gyro aligned with the second body axis of the aerial vehicle) is approximately 0.6 deg/s for most of

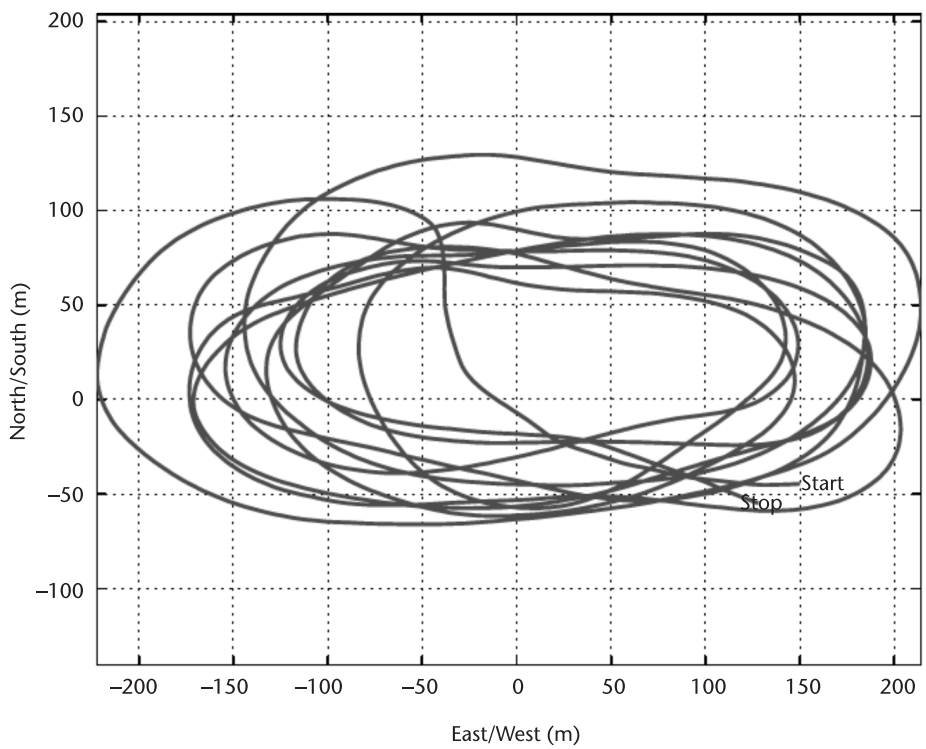

Figure 7.1 Ground trajectory of a small flying wing described in [2].

1. The sensor suite used was the low-cost GPS/INS system known as the Micro Nav produced by Crossbow. At the time of the writing of this chapter, Crossbow announced the discontinuation of the Micro Nav product. However, the performance of the inertial sensors that made up the Micro Nav are typical of many low-cost IMUs. Thus, using another low-cost IMU will lead to more or less similar results.

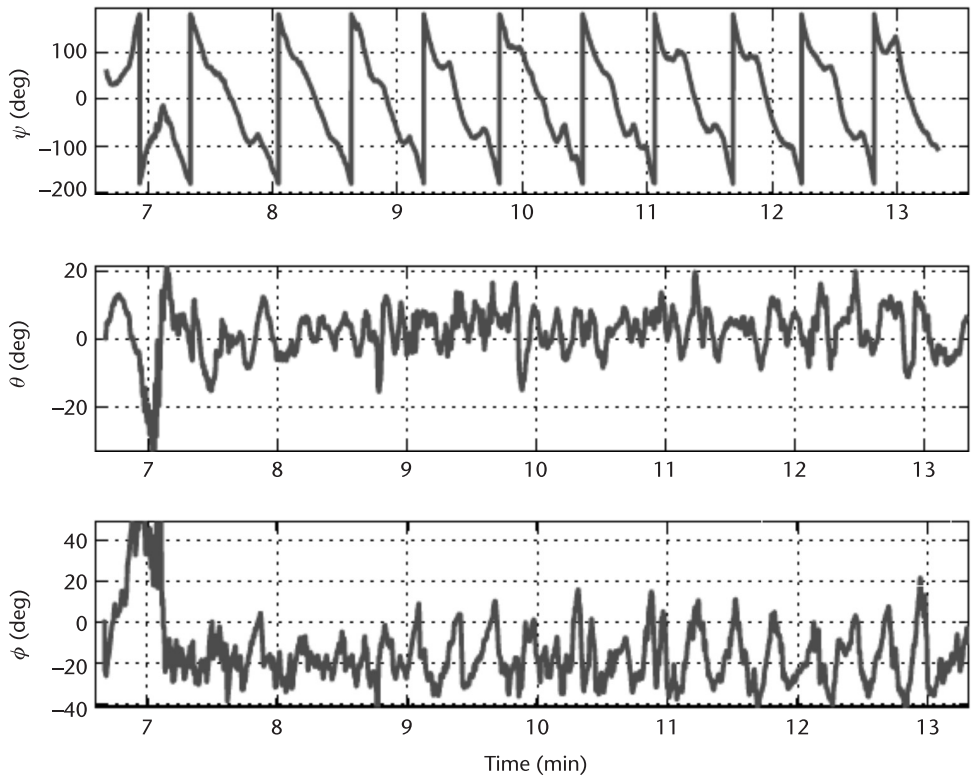

Figure 7.2 Attitude history for a low-cost GNSS/INS integration.

the time. This is the combined effect of null shift and time varying biases. Thus, without any aiding, the attitude drift due this rate gyro could be as large as 36 deg/min.

Returning back to the error estimates of Figure 7.3, we note that a drift rate of 0.6 deg/s is large in the sense that would lead to a rapid error growth in the navigation state vector estimate without the GNSS aiding. To see the effect of a GNSS outage, we examine the velocity history for the same flight test. Figure 7.4 shows the estimate of the east component of the velocity vector as generated by the GNSS/INS algorithm. We compare this to the GPS generated estimate of v_E^n. The GNSS aiding is turned off at t = 10 minutes for a duration of 30 seconds. From Figure 7.4 we see that without GNSS aiding the velocity solution drifts very quickly. The behavior of the GNSS/INS algorithm seen in Figure 7.4 gives credence to the statement we made in Chapter 6 that the navigation solution from a low-cost inertial sensor suite is practically useless without GNSS aiding.

Note that in Figure 7.4 when GNSS aiding is restored at t = 10.5 minutes, the GNSS/INS velocity solution starts converging back to the stand-alone GPS velocity estimate. The rate of convergence is such that it requires approximately 0.1 minutes (or 6 seconds) for the velocity estimates to stabilize. This rate of convergence can be affected by the appropriate "tunning" of the filter entries in the process noise covariance matrices (i.e., changing the values). Care must be taken when doing this, however, as large transients that may result during the convergence period

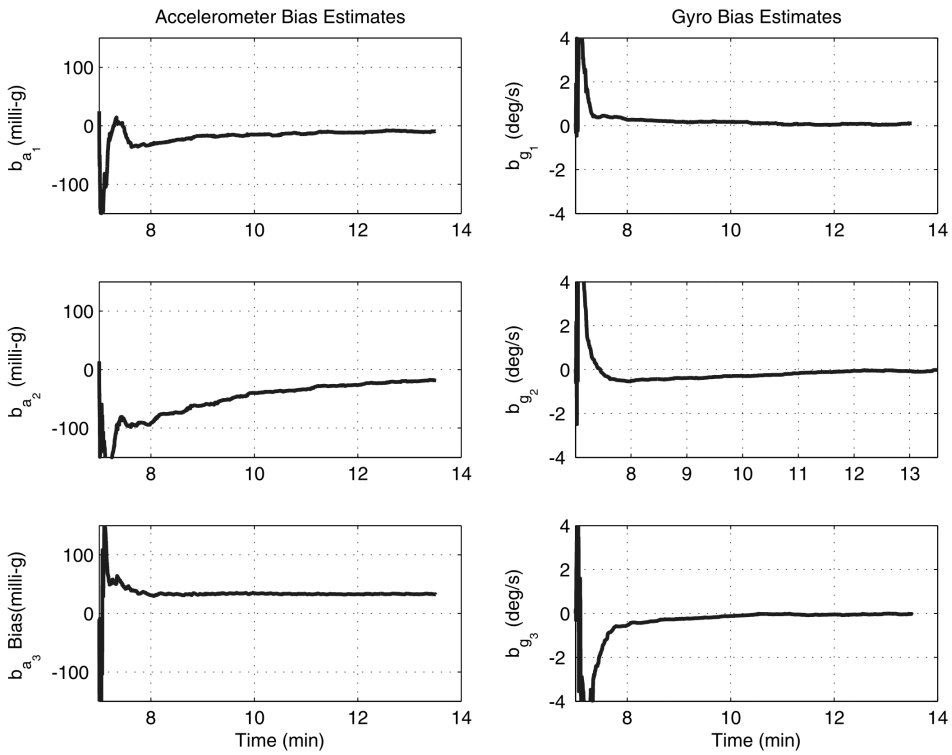

Figure 7.3 Sensor bias estimate for the low-cost GNSS/INS.

may destabilize the filter and lead to divergence (i.e., estimation errors growing without bound).

Figure 7.5 is a "zoom in" of the GNSS/INS algorithm position estimate. In particular, it shows the estimate of x^e (x-component of the vehicle's position in ECEF coordinates). It can be seen that the GPS position estimates are periodic and appear once every second (1-Hz update rate). On the other hand, the GNSS/INS algorithm's output occurs at a rate of 50 Hz. We show Figure 7.5 to highlight that one of the important things an IMU brings to the integrated GNSS/INS solution is higher bandwidth of information in small platforms such as UAVs. It is particularly important that high-bandwidth attitude information be available because it is indispensable for automatic control. Thus, without the integration of GNSSs and INSs, it would be very difficult to implement autopilots for small UAVs.

7.3 Case Study 2: Vehicle Sideslip Estimation

From the discussions associated with case 1 it is clear that an integrated GNSS/INS algorithm can provide a reasonably accurate, three-axis attitude estimate. Thus, one may conclude that multiantenna GPS attitude determination systems such as the ones described in [3, 4] or ad hoc attitude estimators such as those in [5, 6] are not useful. However, there are instances where the GNSS/INS integration as

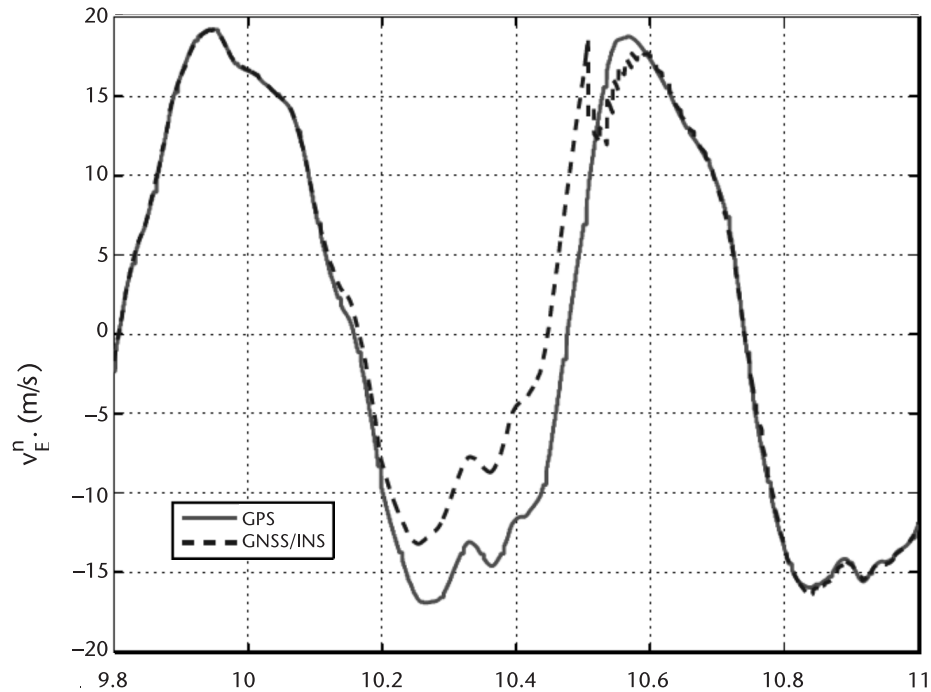

Figure 7.4 Effect of a momentary GNSS outage on the velocity estimate from the GNSS/INS integration algorithm.

described in Chapter 6 (or case 1) cannot provide a three-axis attitude estimate. We will examine this aspect of GNSS/INS integration by considering another application where low-cost inertial sensors and GNSSs are making a large impact—automotive navigation, guidance, and control. In this regard, this case study examines the motivation for using as well as the limitations of a low-cost integrated GNSS/INS systems used to estimate vehicle sideslip as well as critical tire parameters.

7.3.1 Motivation

The sideslip angle (denoted as β here) is the angle between the longitudinal (front-aft) axis of the vehicle and the velocity vector of its center of gravity (CG). Sideslip estimation is a critical aspect of many ESC systems [7–10].

Sideslip estimation schemes fall into two categories, model-based and kinematic. Model based estimators use a dynamic lateral vehicle model such as the bicycle model [11], while kinematic estimators use the kinematic relationships between the measurements of different sensors such as GPS and IMUs [12, 13]. Model-based estimators are typically cleaner than kinematic estimators. However, they require knowledge about the vehicle parameters that is often difficult to obtain (such as cornering stiffness). The only vehicle-dependent knowledge kinematic estimators require is the location of the sensors and CG to translate velocities and accelerations to a common point using something akin to (6.46) in the previous chapter. However, even this is often neglected.

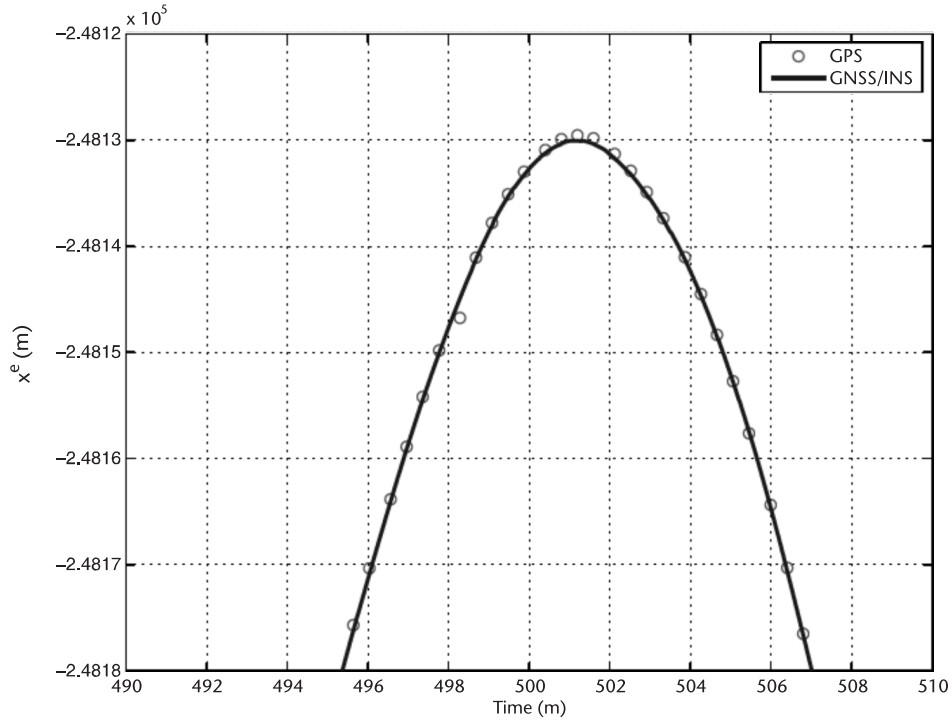

Figure 7.5 Close-up of the ECEF (or Earth frame) *x* coordinate estimate.

A typical method to measure sideslip on test vehicles is using an optical sensor to measure the lateral and longitudinal velocity. However, these sensors are expensive and their performance can be dependent on the surface on which the vehicle is traveling. For these reasons, on production vehicles, this state is not measured. A newer and inexpensive (relative to the optical sensors) way to measure sideslip is using a low-cost integrated navigation system such as the GNSS/INS systems discussed earlier.

The sideslip angle can be estimated as the difference between the vehicle heading and direction of travel (course) measured at the CG. That is:

$$\beta = \tan^{-1}\left(\frac{v_E^n}{v_N^n}\right) - \psi_{nb} \tag{7.1}$$

The first term on the right-hand side of (7.1) can be computed using the outputs of a GNSS receiver or the velocity estimates generated by an integrated GNSS/INS navigator. The second term, ψ_{nb}, is part of the attitude estimate generated by a GNSS/INS navigator. Thus, as long as attitude and velocity estimates are available, sideslip can be estimated. Figure 7.6 shows the outputs of a sideslip estimator. The figure on the left is the sideslip history for a vehicle traveling on a low-friction surface ("low μ") and the one on the right is for asphalt.

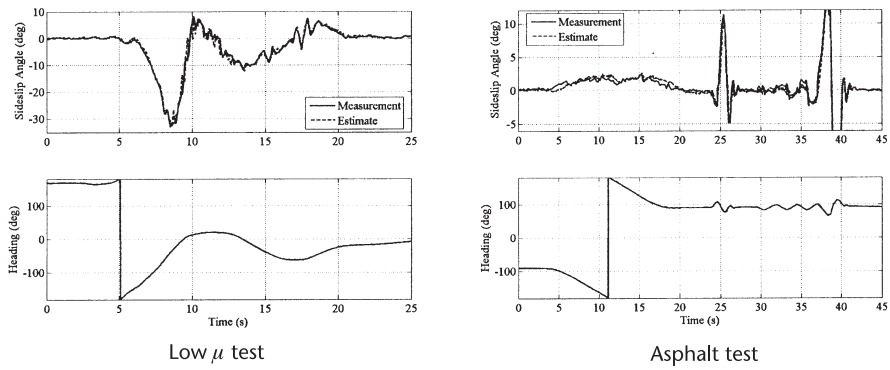

Figure 7.6 Sideslip estimates using a filter-blending GNSS, INS, and a vehicle dynamic model.

These estimators are Kalman filters, which fuse the information from a vehicle dynamic model with GNSSs and INSs. Details on these estimators can be found in [14].

The ultimate purpose of sideslip estimation is to predict the cornering stiffness and peak lateral force of the front and rear axles by estimating the lateral force and tire slip angle over the entire operating range. For example, the lateral force estimates generated for an Infiniti G-35 Sedan using sideslip information are shown in Figure 7.7.

Unfortunately, high-quality estimates of β are not always available. This occurs when GNSS measurements are not available, the observability of the ψ_{nb} estimate is poor, or both. The former occurs, for example, in urban canyons where the GNSS signals are weak or obstructed. The latter occurs when the GNSS information is not sufficient to keep errors stemming from inertial sensor biases in check. This latter case is related to the issue of observability, which we will discuss in some detail in Section 7.3.2.

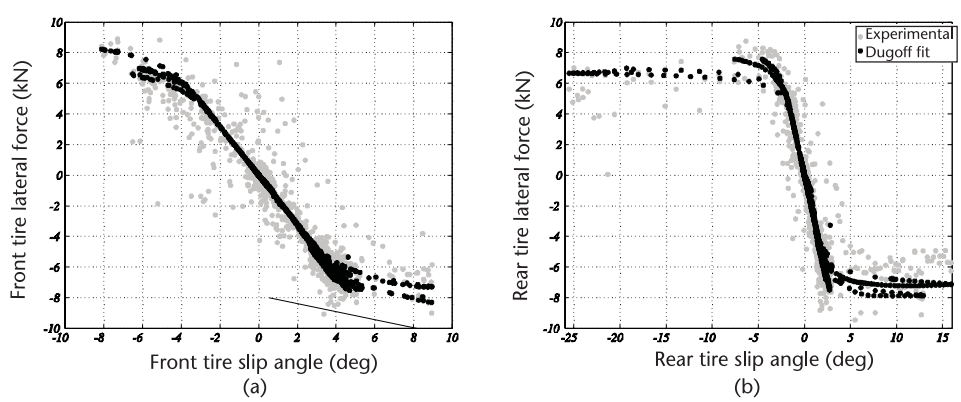

Figure 7.7 Lateral force estimates for an Infiniti G-35 sedan [14]. The output of an INS/GNSS integrated navigator is an input to the algorithm, which estimates these side forces.

7.3.2 Observability

Information from GNSSs may not be able to keep inertial sensor errors in check when one of the following situations arise:

1. Information from GNSS does not have a "path" to get to the inertial sensors and calibrate them. This is a condition known as lack of *geometric* or *deterministic* observability [15–21].
2. Information from the GNSS has a "path" to get to the inertial sensor but cannot arrest the error growth rate of the inertial solution. This is said to be a lack of stochastic observability [16, 18].

In what follows, we will make clear what is meant by "path from GNSS to inertial sensors." The important point to take away from this discussion is that there are conditions under which the integration of GNSSs and inertial sensors will not result in a drift-free attitude solution. This point can be made clearer by considering the specific example of sideslip estimation under the conditions shown in Figure 7.8.

Figure 7.8 shows two different trajectories that are traveled by an automobile at constant speed. In both cases, the vehicles are traveling north, and we invoke the flat-Earth assumptions. That is, the vehicle is moving perfectly level (i.e., h = constant and $\dot{v}_D = 0$) and we are considering short propagation times and motion

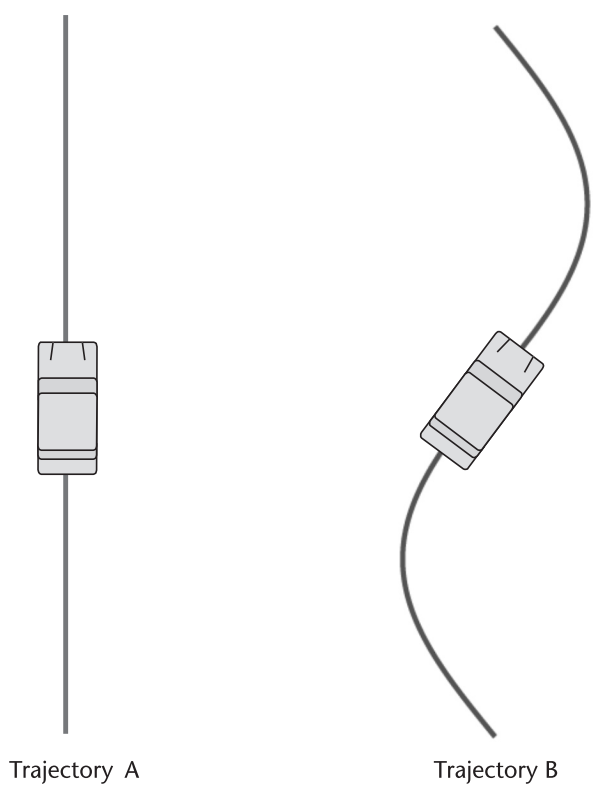

Figure 7.8 Hypothetical vehicle trajectories.

over short distances. The short propagation times enable us to ignore the complexities associated with a curved Earth (which come through the matrix **T**) as well as ensuring that we do not fall off the hypothetical flat Earth. Since we are assuming a flat Earth and constant speeds, during the constant heading trajectory A ($\psi = 0°$) the vehicle will not experience acceleration. Because of the continually changing heading along trajectory B, however, the vehicle will experience an acceleration. We will consider the case where the inertial sensors used are of the automotive- or consumer-grade quality. Given these approximations, as we will show next, for trajectory A, the heading errors resulting from gyro drift cannot be arrested by GNSS position and velocity measurements while, on trajectory B, they can. This is because on trajectory A, heading lacks observability. To see why heading is unobservable on trajectory A and not trajectory B, let us examine the velocity differential equation. Since the gyro output biases are large and we are assuming a flat Earth, we can write the following equation:

$$\dot{\mathbf{v}} \approx \mathbf{C}_b^n \mathbf{f}^b + \mathbf{g} \tag{7.2}$$

Since the vehicle is perfectly level ($\theta = \phi = 0$), then referring to Figure 7.9 the north and east components of acceleration are:

$$a_N = \dot{v}_N = f_x \cos\psi - f_y \sin\psi \tag{7.3}$$

$$a_E = \dot{v}_E = f_x \sin\psi + f_y \cos\psi \tag{7.4}$$

For the moment if we assume the only imperfect sensor is the gyro measuring rotation about the body z-axis of the vehicle, then the heading angle estimated by the inertial sensors, $\hat{\psi}$, will be related to the true heading, ψ, by:

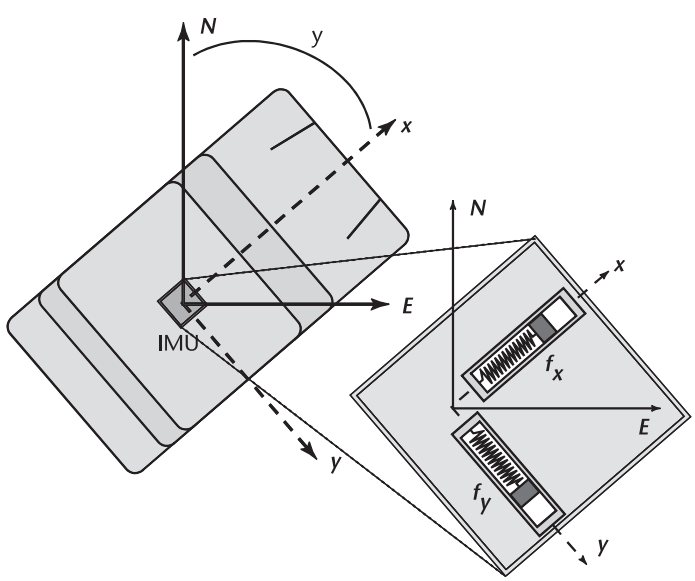

Figure 7.9 Sensor axes arrangement relative to the navigation frame.

$$\hat{\psi} = \psi + \delta\psi = \psi + \int_{t_0}^{t} b_{g_3}\, d\tau \tag{7.5}$$

where b_{g_z} is the z-axis rate gyro bias. Assuming that $\delta\psi$ is small allows dropping higher order terms when we substitute the above expression for $\hat{\psi}$ into (7.3) and (7.4). This gives:

$$\hat{a}_N = a_N + (f_x \cos\psi + f_y \sin\psi)\ \delta\psi = a_N + \delta a_N \tag{7.6}$$

$$\hat{a}_E = a_E + (f_x \sin\psi + f_y \cos\psi)\ \delta\psi = a_E + \delta a_E \tag{7.7}$$

The variables $\hat{a}_N$ and $\hat{a}_E$ represent the acceleration quantities that will be used by the inertial navigation algorithm to compute a velocity and position estimate. If the vehicle is not experiencing acceleration, f_x and f_y will be zero. Thus, $\hat{a}_N = a_N$ and $\hat{a}_E = a_E$, meaning that any heading errors resulting from gyro drift will not be reflected in the velocity and position estimates generated by using the inertial sensor outputs. This means that there will be no error signal (i.e., no difference between the GNSS and INS estimated states), which means that $\delta\mathbf{y}_k$ in (6.32) is zero and, thus, no corrections will be computed (i.e., $\delta\mathbf{x}_k = 0$). In this case, the heading error can grow without check.

Of course, any realistic maneuver will involve even slight accelerations. Thus, heading errors will *always* affect the velocity and position estimates. However, if the accelerations are small the EKF error signal $\delta\mathbf{y}_k$ will be small and very few corrections will be calculated. So, if the rate gyro biases are large, this can lead to heading errors that grow at a rate that cannot be arrested by the GNSS measurements. This, in turn, can lead to EKF divergence (i.e., errors growing without bounds and/or state estimates that do not converge to the correct value).

While the two examples above rely on low-cost inertial sensors, the following case study considers high-accuracy applications that often use higher quality inertial sensors.

7.4 Case Study 3: INS To Aid High-Accuracy GNSSs

For many applications, the meter-level positioning accuracy obtained when using GNSS pseudorange measurements is insufficient. These applications rely instead on carrier-phase measurements, which, as will be discussed in Chapter 14, provide highly precise ranging capability. If the carrier phase ambiguities can subsequently be resolved to their integer values, positioning accuracy at the centimeter level is possible. Unfortunately, resolving the ambiguities to their integer values is not a trivial task. This case study looks at how inertial navigation can help to improve the speed and reliability of the ambiguity-resolution process. However, as will be shown later, this requires an inertial navigation solution with exceptional accuracy. The low-cost inertial sensors and associated mechanizations that have thus far been the focus of this chapter will not satisfy the performance required for this application.

7.4.1 GNSS Ambiguity-Resolution Overview

A full understanding of ambiguity resolution is beyond the scope of this case study, but a cursory overview will help readers to understand the fundamental concepts. In this spirit, we consider a GNSS-only Kalman filter that uses double-difference (see Chapter 14 for details on double differencing) pseudorange and carrier-phase measurements to estimate the position states and the double-difference carrier-phase ambiguities. We write the state vector as:

$$\mathbf{x}^T = \begin{bmatrix} \mathbf{p}^T & \mathbf{N}^T \end{bmatrix} \tag{7.8}$$

where $\mathbf{p}$ is the position vector and $\mathbf{N}$ is the vector of double difference ambiguities. The corresponding covariance matrix of the above states is denoted in hyper-matrix form as:

$$\mathbf{P} = \begin{bmatrix} \mathbf{P}_{p,p} & \mathbf{P}_{p,N} \\ \mathbf{P}_{N,p} & \mathbf{P}_{N,N} \end{bmatrix} \tag{7.9}$$

where $\mathbf{P}_{a,b}$ is the covariance matrix between vectors a and b. The covariance matrix of the ambiguities, $\mathbf{P}_{N,N}$, plays an important role in ambiguity resolution. Specifically, it has been shown in [21] that $\mathbf{P}_{N,N}$ can be used to compute the ambiguity DOP (ADOP), which is related to the volume of the hyperellipsoid defined by the ambiguity covariance matrix (centered at $\mathbf{N}$). The volume of the hyperellipsoid is important because it determines the size of the *search space* within which the true ambiguity vector is assumed to lie. All possible ambiguity combinations within the search space are then tested to select the best set, which is then considered as correct. Generally speaking, the smaller the search space, the faster and more reliable the ambiguity resolution process will be.

With this in mind, the GNSS-only approach to ambiguity resolution *conceptually* uses the pseudorange measurements to estimate the position. The carrier-phase measurements then use the estimated position to derive an estimate of the ambiguities with a commensurate level of accuracy. As the filter receives more GNSS measurement updates, the position estimates get better determined (until steady state), which reduces the uncertainty in the estimated ambiguities, thus reducing the search space and improving ambiguity resolution performance. Stated differently, the reason that the instantenous ambiguity resolution is difficult is because of the inaccuracy of the pseudorange measurements relative to the carrier-phase wavelength.

7.4.2 Benefits of INSs to Ambiguity Resolution

The use of an INS can benefit the ambiguity-resolution process in two ways. First, in a tight integration configuration, the INS reduces the amount of process noise in the system model compared to the GNSS-only case. In turn, this allows for a more rapid convergence of the filter and lower steady-state covariance. This benefit is effectively realized continuously because of the nature of how the system is set up and is therefore not discussed further. The second benefit of the INS occurs

following an interuption (outage) of GNSS measurements—for example, after driving through a tunnel. This benefit can be realized using a loose or a tight integration approach [22, 23] and is discussed in more detail here.

After a GNSS data outage, the carrier-phase amguities estimated in the Kalman filter must be reset and the convergence process described in Section 7.4.1 be restarted. Conceptually then, in a GNSS/INS system the inertial position provides additional information that is unavailable with GNSSs alone, and one would therefore expect an improvement in ambiguity-resolution performance. To this end, investigations into the benefit of INS for high-accuracy positioning include, for example, [22, 27]. From [26, 27], assuming that only carrier-phase measurements are used to update the filter (for mathematical simplicity only), the ambiguity covariance matrix, $\mathbf{P}_{N,N}$, will take the following form after the first GNSS update following a GNSS outage

$$\mathbf{P}_{N,N} = \mathbf{P}_{N,N} - \lambda^2 \mathbf{P}_{N,N}\left(\mathbf{D}^T \mathbf{H} \mathbf{P}_{p,p} \mathbf{H}^T \mathbf{D}_{N,N} + \lambda^2 \mathbf{P}_{N,N} + \mathbf{R}_\phi\right)^{-1} \mathbf{P}_{N,N} \tag{7.10}$$

where λ is the wavelength of the carrier-phase measurement, $\mathbf{H}$ is the design matrix for the undifferenced GNSS measurements, $\mathbf{D}$ is the double difference coefficient matrix, and $\mathbf{R}_\phi$ is the covariance matrix of the carrier phase measurements. We draw attention to the first term in brackets, which is proportional to the position covariance matrix following the outage. The smaller the covariance (in a matrix sense), the larger will be the term being inverted, and consequently, the larger the term being subtracted from the initial ambiguity covariance matrix (which is usually initialized to be large in order to reflect the unknown value of the ambiguities). Herein lies the benefit of the inertial data. Specifically, if the INS can bridge the GNSS data outage with sufficient accuracy then the ambiguity covariance matrix can be reduced relative to the GNSS-only case, resulting in improved ambiguity resolution. Although there is no hard and fast rule for defining "sufficient accuracy" in this context, [22, 23] show that once the INS accuracy degrades near that of the GNSS code solution accuracy, little benefit is gained.

Finally, it is noted that the above development said nothing of the quality of the IMU required. However, it is realized that higher quality sensors will inherently provide better INS accuracy over a given time interval (GNSS outage duration). Alternatively, for a higher quality IMU the duration over which GNSS data is unavailable can be extended relative to the case when a lower cost unit is used, with the same benefit.

7.5 Software Examples

One cannot appreciate the subtleties associated with GNSS/INS integration and inertial-aided carrier phase differential GNSS integer ambiguity resolution from the foregoing simple discussion. Thus, we urge the reader interested in gaining a deeper understanding of this topic to refer to the various papers and books listed at the end of this chapter. We also encourage the readers to start exercising the MATLAB-based simulations that accompany this book.

The MATLAB examples that are provided are:

1. An extended Kalman filter implementation of a loose GNSS/INS integration discussed in Section 7.2. Data from an actual IMU as well as data generated by a simulator are provided. The reader can play back these data sets and explore various aspects of GNSS/INS integration.
2. A 2D GNSS/INS integrated extended Kalman filter that demonstrates the observability analysis discussed in Section 7.3.
3. ADOP analysis examples to show the performance required by inertial navigation algorithms if they are to be used in carrier phase integer ambiguity aiding.

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